Modeling BTC Order Flow with Hawkes Processes:

Implications for Pricing, Hedging, and AI-Driven Execution

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Abstract

This paper builds a framework for modeling Bitcoin (BTC) order flow using Hawkes self-exciting point processes and applies the resulting real-time intensity signal $\lambda(t)$ across four interconnected quantitative finance problems. Starting from stochastic calculus foundations—Brownian motion, Itô's lemma, jump-diffusion, and Lévy processes—we calibrate a Hawkes process to 80,000 BTC-USD tick trades from Coinbase (May 2026), obtaining branching ratio $\alpha/\beta = 0.185$, confirming stationarity. The Hawkes-Vol model $\sigma^2(t) = \sigma^2_{\text{base}} + \gamma\lambda(t)$ achieves 66.3% directional accuracy in realized volatility forecasting, outperforming GARCH(1,1). A Hawkes+NIG option pricing model reduces ATM implied volatility from 40.25% (Black-Scholes) to 20.40% via Carr-Madan FFT, reproducing the left-skewed IV smile of BTC markets. Embedding $\lambda(t)$ in DRL state vectors yields +868% improvement in delta hedging PnL (PPO: +\$361.3), +99.2% in optimal execution reward (SAC vs. TWAP), and −21% inventory risk in market making. These results show that Hawkes intensity is a useful real-time signal connecting tick-level microstructure and macro financial applications.

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1. Introduction

2. Introduction

Quantitative methods have become increasingly important in finance. Early work on insurance and life tables built the first links between probability and financial decisions. Later, the growth of commodity markets and structured products led to new pricing theories. The Black–Scholes–Merton model is the most famous result from that period. But the 2008 global financial crisis showed the weaknesses of those classical models. At the same time, it pushed researchers to look more closely at how prices are actually formed. With the computer technology improvement, the focus shifted from *what prices should be* to *how prices come about*.

Dr.Maureen O'Hara. Her work on market microstructure set up a central idea: prices move not only because of public news, but also because of who is trading, how large the orders are, the direction of orders, and what market makers infer from the order flow. Around the same time, Almgren and Chriss (**AlmgrenChriss2001**) showed that optimal execution is a stochastic control problem: your own trades push the price, so you must balance speed and market impact. Later, Avellaneda and Stoikov (**AvellanedaStoikov2008**) framed market making as a dynamic control problem, using Hamilton–Jacobi–Bellman equations to jointly handle inventory risk and bid-ask quotes. After these works, quantitative trading became a mix of stochastic control, differential equations, and dynamic optimization.

Many of these early models assumed that orders arrive independently, like a Poisson process. But real order flow shows strong clustering and chain reactions: one trade tends to bring more trades. The Hawkes process is the natural tool for this self-exciting behavior. Its intensity depends on past events, so it captures clustering in a direct way (**Hawkes1971**; **Bacry2015**). Because of this, the Hawkes process builds a clean bridge between microstructure, order flow, point processes, and optimal control:

$$\text{Microstructure} \;\rightarrow\; \text{Order Flow} \;\rightarrow\; \text{Point Process} \;\rightarrow\; \text{Optimal Control.}$$

The story does not stop at order-flow modeling. Hawkes-type clustering in orders leads to volatility clustering. Option pricing, from Black–Scholes to more complex jump models, is very sensitive to changes in volatility. As a result, market makers and high-frequency traders need to update their Greek exposures—especially Delta, Gamma, and Vega—based on the current order-flow intensity. In today's electronic markets, many strategies boil down to managing three things at the same time:

$$\text{Inventory Risk} \;+\; \text{Market Impact} \;+\; \text{Greek Exposure.}$$

Because of this, microstructure, volatility modeling, asset pricing, and Greek management can no longer be treated as separate topics. They have slowly come together into one unified trading framework. This paper sits right at that intersection.

We chose Bitcoin (BTC) as our testing ground, and the choice is not random. BTC trades 24 hours a day, 7 days a week, on electronic platforms without designated market makers. It shows order-flow clustering in a stronger form than most stock markets. In our dataset, the autocorrelation of trade signs reaches 0.42 over a five-hour window—a level not often seen in liquid equity markets—and the average trade rate of 4.13 trades per second gives us a rich, high-frequency series for reliable estimation. BTC therefore is a good laboratory for testing whether the Hawkes framework works in practice across different trading tasks.

In this paper, we ask whether the Hawkes conditional intensity $\lambda(t)$, a single number extracted only from trade arrival times, carries useful information for several financial problems. We estimate a univariate Hawkes process from BTC-USD tick data, and then use the resulting intensity in volatility forecasting, option pricing, delta hedging, and algorithmic trading. The results, described in detail in the main sections, point to a simple message: a tick-level point-process signal can serve as a flexible input that connects microstructure to the pricing and execution layers of quantitative finance.

The rest of the paper is organized as follows. Section 2 develops the theoretical background, moving from Brownian motion and Itô's lemma through jump processes and Lévy processes to the Hawkes process. Section 3 describes the data and key microstructure statistics. Section 4 presents the maximum likelihood estimation and goodness-of-fit tests. Sections 5 to 7 cover the three applications: volatility forecasting, option pricing and Greeks, and deep reinforcement learning for trading. Section 8 discusses limitations and future work. Section 9 concludes.

3. Theoretical Foundations

This section builds the math we need, step by step: *continuous diffusion* $\to$ *jumps* $\to$ *Lévy processes* $\to$ *Hawkes process*. Each step adds one feature the previous model was missing. We end with the Hawkes process, which we use to model BTC order flow.

3.1 Brownian Motion and the Itô Integral

Definition 3.1 (Standard Brownian Motion). *A stochastic process* $\{W_t\}_{t\geq 0}$ *is a* standard Brownian motion *if*

1. $W_0 = 0$ *a.s.;*
2. *for any* $0 = t_0 < t_1 < \cdots < t_n$*, the increments* $W_{t_1} - W_{t_0}, \ldots, W_{t_n} - W_{t_{n-1}}$ *are mutually independent;*
3. $W_t - W_s \sim \mathcal{N}(0, t-s)$ *for all* $0 \leq s < t$*;*
4. *paths* $t \mapsto W_t$ *are continuous a.s.*

In financial modeling, W_t represents the continuous, small, unpredictable fluctuations of market noise. Because W_t is nowhere differentiable, classical calculus does not apply to integrals of the form $\int_0^T \Delta_s \, \mathrm{d}S_s$ (the profit from a hedging strategy). This motivates the *Itô integral*:

Definition 3.2 (Itô Integral)**.** *For an adapted process* $\{X_t\}$*, the Itô integral is* $I_t = \int_0^t X_s \, \mathrm{d}W_s$*. It satisfies the* martingale property*:* $\mathbb{E}[I_t] = 0$*.*

3.2 Geometric Brownian Motion and Itô's Lemma

The classical asset-price model assumes S_t follows *geometric Brownian motion* (GBM):

$$\mathrm{d}S_t = \mu S_t \, \mathrm{d}t + \sigma S_t \, \mathrm{d}W_t, \tag{1}$$

where μ is the drift and $\sigma > 0$ is the constant volatility. The discrete-time analogue is the *lognormal random walk*:

$$\ln S_{t+\Delta t} = \ln S_t + \mu \Delta t + \sigma \epsilon \sqrt{\Delta t}, \qquad \epsilon \sim \mathcal{N}(0,1), \tag{2}$$

whose continuous-time limit recovers (1).

Theorem 3.1 (Itô's Lemma)**.** *For* $f(t, S_t)$ *twice continuously differentiable and* $\mathrm{d}S_t = a_t \, \mathrm{d}t + b_t \, \mathrm{d}W_t$*:*

$$\mathrm{d}f = \left(\frac{\partial f}{\partial t} + a_t \frac{\partial f}{\partial S} + \tfrac{1}{2} b_t^2 \frac{\partial^2 f}{\partial S^2} \right) \mathrm{d}t + b_t \frac{\partial f}{\partial S} \, \mathrm{d}W_t. \tag{3}$$

The second-order term $\frac{1}{2} b_t^2 \partial^2 f / \partial S^2$ arises from $(\mathrm{d}W_t)^2 = \mathrm{d}t$ and has no classical analogue. Applying (3) to $f(S_t) = \ln S_t$ with $f' = 1/S$, $f'' = -1/S^2$ gives

$$\mathrm{d}(\ln S_t) = \left(\mu - \tfrac{\sigma^2}{2} \right) \mathrm{d}t + \sigma \, \mathrm{d}W_t, \tag{4}$$

so the explicit solution of (1) is

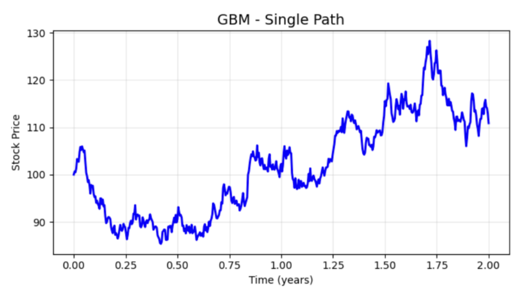
$$S_t = S_0 \exp\left[\left(\mu - \tfrac{\sigma^2}{2} \right) t + \sigma W_t \right]. \tag{5}$$

Applying (3) to a derivative $V(t, S_t)$ and constructing a delta-neutral portfolio yields the *Black-Scholes PDE*:

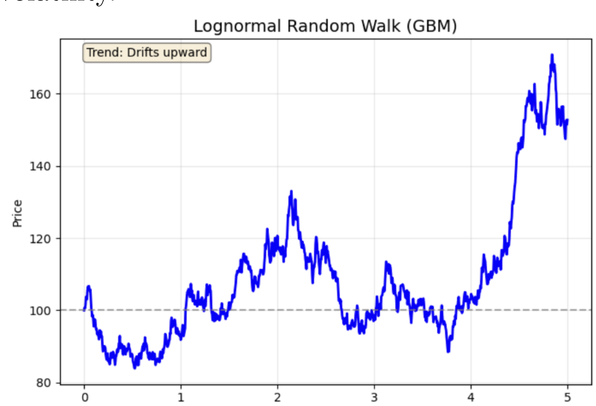
$$\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0, \tag{6}$$

with European call solution $C = S_0 N(d_1) - Ke^{-rT}N(d_2)$, $d_1 = [\ln(S_0/K) + (r+\sigma^2/2)T]/(\sigma\sqrt{T})$, $d_2 = d_1 - \sigma\sqrt{T}$.

Figure 1a illustrates a single GBM path with upward drift ($\mu = 0.05$, $\sigma = 0.2$, $T = 2$). The lognormal random walk discrete approximation is shown in Figure **??**.



(a) GBM single path ($S_0 = 100$, $\mu = 0.05$, $\sigma = 0.2$, $T = 2$). Upward drift with multiplicative volatility.



(b) Logarithmic random walk path with corresponding parameters.

Figure 1: Geometric Brownian Motion and Logarithmic Random Walk simulations. The multiplicative structure ensures positivity but imposes constant volatility—a key empirical limitation for asset classes like BTC.

3.3 Mean-Reverting Processes: Ornstein-Uhlenbeck

While GBM has no tendency to return to a long-term level, the *Ornstein-Uhlenbeck* (OU) process introduces mean reversion:

$$\mathrm{d}X_t = \theta(\mu - X_t)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t, \tag{7}$$

where $\theta > 0$ is the speed of mean reversion, μ is the long-term mean, and σ is volatility. The discrete-time Euler scheme is

$$X_{t+\Delta t} = X_t + \theta(\mu - X_t)\Delta t + \sigma\sqrt{\Delta t}\,\epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0,1). \tag{8}$$

The conditional moments are: The conditional moments of the Ornstein–Uhlenbeck process are given by

$$\mathbb{E}[X_{t+\Delta t} \mid X_t] = \mu + (X_t - \mu)e^{-\theta\Delta t}, \tag{9}$$

$$\mathrm{Var}[X_{t+\Delta t} \mid X_t] = \frac{\sigma^2}{2\theta}(1 - e^{-2\theta\Delta t}), \tag{10}$$

with stationary distribution $\mathcal{N}(\mu,\, \sigma^2/2\theta)$. Financial applications include the Vasicek interest rate model, stochastic volatility, and pairs-trading spread modeling.

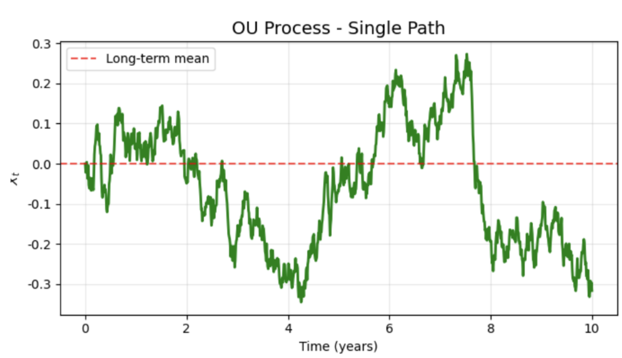


Figure 2: Ornstein-Uhlenbeck process: the path fluctuates but always reverts to long-term mean μ (dashed red). This contrasts with GBM which has no such gravitational pull.

3.4 Why Brownian Motion Falls Short: Jump Processes

GBM makes two assumptions that do not hold for BTC:

1. **Continuous paths**: cannot model sudden price drops (e.g., \$100 $\to$ \$90 instantaneously).
2. **Normal increments**: cannot capture the fat tails observed in real data (excess kurtosis = 32.96 in our BTC dataset).

Figure 3 illustrates both limitations: the top panel contrasts a continuous BM path with a path containing instantaneous $\pm\$15$ jumps, while the bottom panel shows the Normal distribution's inability to fit the fat-tailed empirical return distribution.

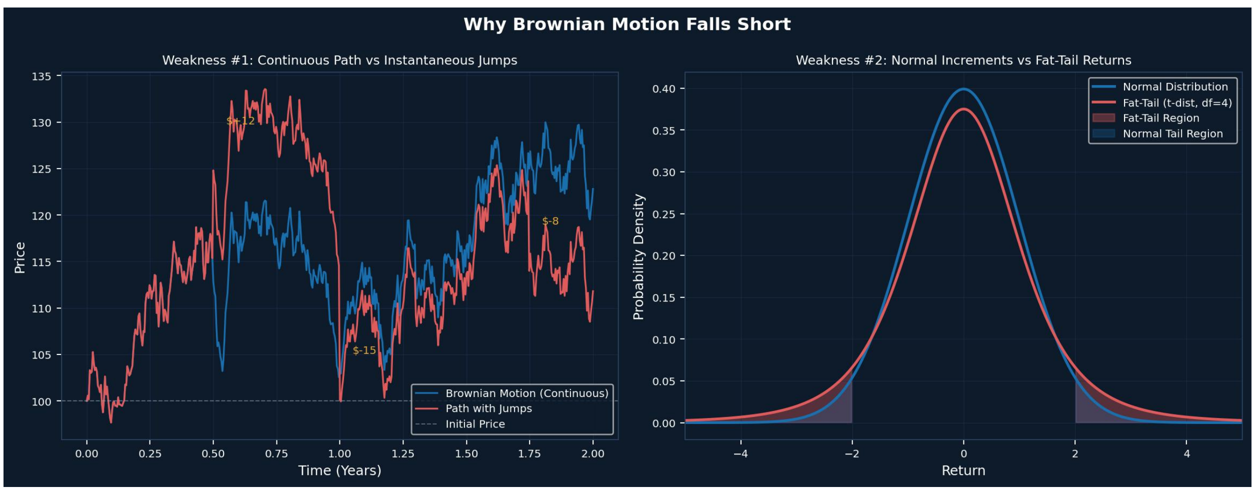


Figure 3: Two weaknesses of Brownian motion. *Top*: continuous BM path (blue) vs. path with instantaneous jumps (red). *Bottom*: Normal distribution (blue) vs. fat-tailed t-distribution (red); extreme events are severely underweighted by the Gaussian model.

Point processes and counting processes. A *point process* records the *times* at which events occur: $0 < T_1 < T_2 < T_3 < \cdots$. Its associated *counting process* is

$$N(t) = \sum_{n=1}^{\infty} \mathbf{1}_{\{T_n \le t\}}, \quad N(0) = 0, \tag{11}$$

counting the number of events up to time t. Figure 4 illustrates the distinction: the point process records the spike times while the counting process integrates them into a piecewise-constant path.

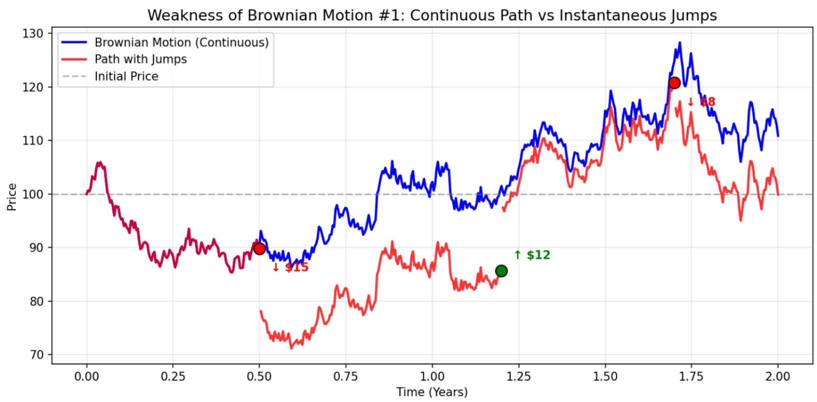


Figure 4: Point process (top: event times as spikes) vs. counting process $N(t)$ (bottom: staircase). This duality underlies the Hawkes framework developed in Section 5.

Poisson process. A *Poisson process* N_t with intensity $\lambda > 0$ satisfies $N_0 = 0$, has independent and stationary increments, and

$$(N_{t+h} - N_t = k) = \frac{(\lambda h)^k}{k!} e^{-\lambda h}, \qquad \mathbb{E}[N_t] = \lambda t. \tag{12}$$

Inter-arrival times are i.i.d. $\mathrm{Exp}(\lambda)$.

Compound Poisson process. Adding random jump sizes $\{J_i\}$ i.i.d. gives the *compound Poisson process*:

$$Y_t = \sum_{i=1}^{N_t} J_i, \tag{13}$$

which stays constant between arrivals and leaps by J_i at time T_i.

Jump-diffusion (Merton 1976). The Merton model combines GBM with compound Poisson jumps:

$$\mathrm{d}S_t = \mu S_{t-}\ \mathrm{d}t + \sigma S_{t-}\ \mathrm{d}W_t + S_{t-}\ \mathrm{d}J_t, \qquad J_t = \sum_{i=1}^{N_t}(Y_i - 1), \tag{14}$$

where $Y_i \sim \mathrm{LogNormal}(\mu_J, \sigma_J^2)$. When a jump occurs, $S_t = Y_i S_{t-}$, creating an instantaneous discontinuous change. Itô's lemma for jump processes adds a jump contribution: $[f(S_{t-}Y_i) - f(S_{t-})]\ \mathrm{d}N_t$. Applying this to $X_t = \log S_t$:

$$\mathrm{d}X_t = \left(\mu - \tfrac{\sigma^2}{2}\right)\ \mathrm{d}t + \sigma\ \mathrm{d}W_t + \log(J)\ \mathrm{d}N_t, \tag{15}$$

so the integrated log-price is the Lévy-type representation $X_t = X_0 + (\mu - \sigma^2/2)t + \sigma W_t + \sum_{i=1}^{N_t} \log J_i$.

Figure 5 shows the three-layer simulation: Poisson counting process → compound Poisson cumulative jump → Merton jump-diffusion asset price.

Figure 5: Simulation hierarchy. *Top*: Poisson process ($\lambda = 2$) counting jump arrivals. *Middle*: compound Poisson cumulative jump size ($J_i \sim \mathcal{N}(0, 0.1^2)$). *Bottom*: Merton jump-diffusion asset price ($\mu = 0.05$, $\sigma = 0.20$, $\lambda = 2$) showing sudden large drops.

3.5 Lévy Processes and the NIG Distribution

Definition 3.3 (Lévy Process)**.** *A càdlàg process $\{X_t\}_{t\geq 0}$ with $X_0 = 0$ is a* Lévy process *if it has (i) independent increments, (ii) stationary increments ($X_{t+h} - X_t \stackrel{d}{=} X_h$), and (iii) stochastic continuity ($\lim_{h\to 0}(|X_{t+h} - X_t| > \epsilon) = 0$ for all $\epsilon > 0$).*

Lévy processes give one framework that connects diffusion and jumps:

$$\text{Brownian motion} \;\subset\; \text{Jump-diffusion} \;\subset\; \text{Lévy processes.}$$

Every Lévy process is *infinitely divisible*: its characteristic function takes the form $\mathbb{E}[e^{iuX_t}] = e^{t\psi(u)}$ (*Lévy–Khintchine*), where the characteristic exponent $\psi(u)$ captures drift, diffusion, and jump structure jointly.

Lévy–Itô decomposition. Every Lévy process decomposes uniquely as three independent parts:

$$X_t = \underbrace{bt}_{\text{drift}} + \underbrace{\sigma W_t}_{\text{diffusion}} + \underbrace{\int_0^t \int_{|z|<1} z\,\tilde{N}(\,\mathrm{d}s,\,\mathrm{d}z)}_{\text{small jumps (compensated)}} + \underbrace{\int_0^t \int_{|z|\geq 1} z\,N(\,\mathrm{d}s,\,\mathrm{d}z)}_{\text{large jumps}}, \tag{16}$$

where $N(\,\mathrm{d}s,\,\mathrm{d}z)$ is the Poisson random measure with Lévy measure $\nu(\,\mathrm{d}z)$, and $\tilde{N} = N - \nu(\,\mathrm{d}z)\,\mathrm{d}s$ is its martingale compensator. The small-jump integral requires compensation because $\int_{|z|<1} |z|\,\nu(\,\mathrm{d}z)$ may be infinite (infinite-activity).

Normal Inverse Gaussian distribution. Among all Lévy processes, the *Normal Inverse Gaussian* (NIG) model (**BarndorffNielsen1997**) is particularly well suited for BTC returns because it can match negative skewness and heavy tails at the same time, while keeping a closed-form characteristic function.

Definition 3.4 (NIG Distribution)**.** *A random variable X follows a NIG distribution, $X \sim \mathrm{NIG}(\alpha, \beta, \delta, \mu)$, if it is a variance-mean mixture of a Gaussian with an Inverse Gaussian mixing distribution. Its characteristic function is*

$$\varphi_{\mathrm{NIG}}(u; \alpha, \beta, \delta, \mu) \;=\; \exp\!\Big(i\mu u + \delta\Big[\sqrt{\alpha^2 - \beta^2} - \sqrt{\alpha^2 - (\beta + iu)^2}\Big]\Big), \tag{17}$$

where $\alpha > 0$, $|\beta| < \alpha$, $\delta > 0$, $\mu \in \mathbb{R}$.

The four parameters have clear economic meanings:

- α controls *tail heaviness*: smaller α gives heavier tails; $\alpha \to \infty$ recovers the normal distribution.

- $\beta \in (-\alpha, \alpha)$ controls *skewness*: $\beta < 0$ produces a left-skewed distribution, matching the negative skewness (-0.31) observed in our BTC tick data.

- $\delta > 0$ is a scale parameter.

- μ shifts the location.

The mean, variance, skewness, and excess kurtosis of $X \sim \mathrm{NIG}(\alpha, \beta, \delta, \mu)$ are:

$$\mathbb{E}[X] = \mu + \frac{\delta\beta}{\sqrt{\alpha^2 - \beta^2}}, \tag{18}$$

$$\mathrm{Var}[X] = \frac{\delta\alpha^2}{(\alpha^2 - \beta^2)^{3/2}}, \tag{19}$$

$$\mathrm{Skew}[X] = \frac{3\beta}{\alpha\sqrt{\delta(\alpha^2 - \beta^2)^{1/2}}}, \tag{20}$$

$$\mathrm{Kurt}[X] = 3\,\frac{1 + 4(\beta/\alpha)^2}{\delta\sqrt{\alpha^2 - \beta^2}}. \tag{21}$$

The NIG characteristic function (42) is the key ingredient for option pricing in Section 7. The Carr–Madan FFT method only requires the characteristic function of $\ln S_T$, so any Lévy model with a known characteristic function—including NIG—admits fast, exact option prices without Monte Carlo simulation.

For our BTC dataset (excess kurtosis 32.96, skewness -0.31) we calibrate $\alpha = 8.0$ and $\beta = -4.0$, giving a left-skewed, heavy-tailed distribution that GBM and even Merton's model cannot reproduce. Table 1 summarises how NIG and the other models in this paper fit within the Lévy framework.

Table 1: Lévy process special cases

Model	**Type**	**Lévy measure**	**Key feature**
BM / GBM	Diffusion	$\nu = 0$	Gaussian, continuous
Compound Poisson	Finite-activity	$\nu(\mathbb{R}) < \infty$	Countable jumps
Merton JD	Jump-diffusion	$\nu(\mathbb{R}) < \infty$	Normal jump sizes
Variance Gamma	Infinite-activity	$\nu(\mathbb{R}) = \infty$	Time-changed BM
NIG	Infinite-activity	$\nu(\mathbb{R}) = \infty$	Closed-form CF, left skew

3.6 From Lévy to Hawkes: The Missing Dependence

The independence gap. All Lévy processes—Poisson, VG, NIG—share the property of *independent increments*: the probability of a jump today is independent of past events. Yet real financial markets exhibit:

- *Volatility clustering*: large moves beget large moves.
- *Event clustering*: trades arrive in bursts, not uniformly.
- *Risk contagion*: one order triggers a cascade of further orders.

These patterns call for a model where the jump *rate* depends on the history of past events.

Definition 3.5 (Hawkes Process). *A point process N_t with event times $\{t_i\}$ is a* Hawkes process *with parameters (μ, α, β) if its conditional intensity satisfies*

$$\lambda(t) = \mu + \sum_{t_i < t} \alpha \, e^{-\beta(t-t_i)}, \qquad \mu, \alpha, \beta > 0. \tag{22}$$

Each past event at time t_i immediately increases the intensity by α (self-excitation) and the effect decays exponentially at rate β. The general form with kernel $\phi(\cdot)$ is $\lambda(t) = \lambda_0 + \int_0^t \phi(t-s)\, \mathrm{d}N(s)$.

- μ — **Baseline Intensity**

  The parameter μ represents the exogenous arrival rate of trades that occur independently of previous market activity. It captures external information flow, such as macroeconomic announcements, uninformed retail trading, and institutional order submissions.

  A larger μ indicates a market with naturally high trading activity even in the absence of self-excitation effects.

- α — **Self-Excitation Amplitude**

  The parameter α measures how strongly one event increases the probability of future events. In financial markets, this reflects the extent to which one trade triggers subsequent trades through mechanisms such as momentum trading, algorithmic execution, or herd behavior.

  A larger α implies stronger clustering and greater endogenous market dynamics.

- β — **Decay Rate**

  The parameter β controls how quickly the excitation effect from a previous event dissipates over time.

  Since the excitation term decays exponentially as

  $$\alpha e^{-\beta(t-t_i)},$$

  a large β implies short-lived excitation and rapid memory decay, whereas a small β produces persistent clustering and longer-lasting market impact.

An important quantity in Hawkes process theory is the branching ratio

$$\frac{\alpha}{\beta},$$

which measures the average number of secondary events triggered by a single event.

Proposition 3.1 (Stationarity and Mean Intensity)**.** *The Hawkes process is stationary iff the* branching ratio $\rho \coloneqq \alpha/\beta < 1$, *with stationary mean* $\bar{\lambda} = \mu/(1-\rho)$.

Proposition 3.2 (Cluster Representation)**.** *The Hawkes process admits a Galton-Watson cluster representation: each immigrant (baseline Poisson arrival) generates offspring with expected size* ρ*. Stationarity (*$\rho < 1$*) ensures each cluster eventually dies out.*

Key insight: Lévy vs. Hawkes. Lévy processes answer *what* the path looks like (diffusion and/or jumps); Hawkes processes answer *why* events cluster. Table 2 contrasts the two frameworks, and Figure 6 visualizes the difference in a financial context.

Table 2: Lévy vs. Hawkes: structural comparison

Feature	**Lévy**	**Hawkes**
Core logic	Independent increments	History-dependent intensity
Jump intensity	Constant or deterministic	Stochastic, self-exciting
Path pattern	Diffusion + independent jumps	Burst clusters
Primary use	Asset price dynamics	Order flow, event timing
Independence	Yes (defining property)	No (past events affect future rate)

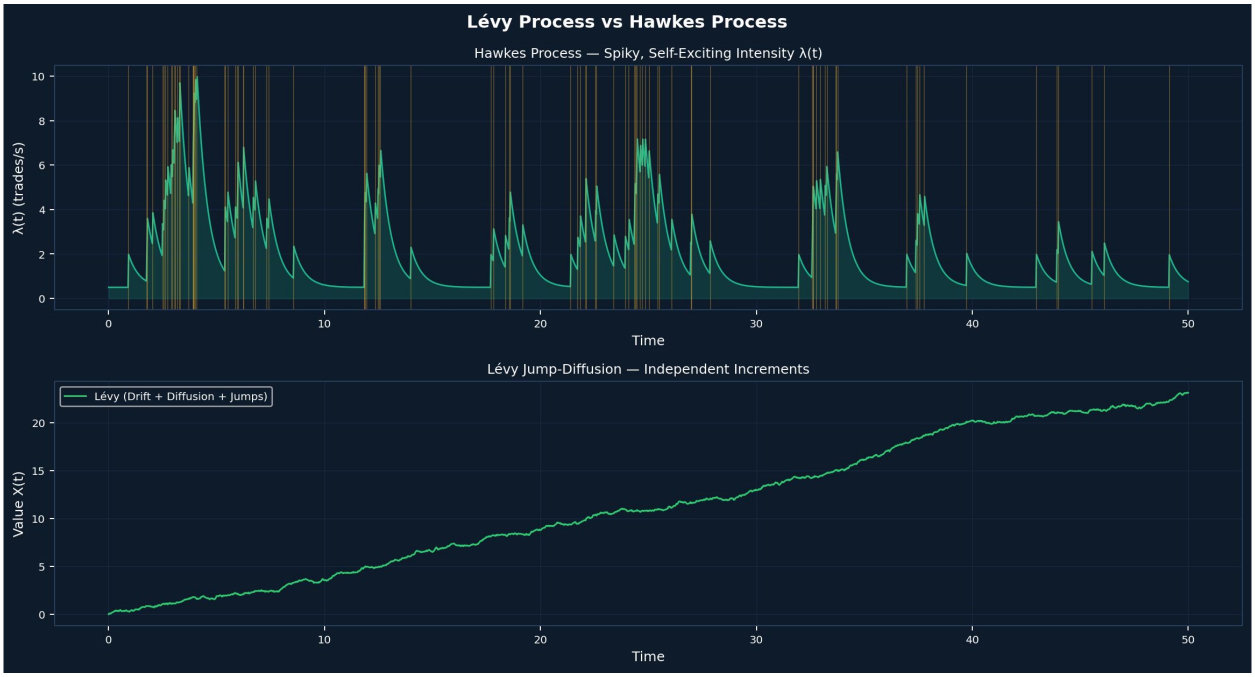


Figure 6: Lévy process vs. Hawkes process. *Top*: Hawkes intensity $\lambda(t)$ (high-excitation regime, $\alpha/\beta = 0.8$) showing spiky, burst-like behaviour. *Bottom*: Lévy jump-diffusion path (drift + diffusion + independent jumps). The Hawkes intensity is stationary while the Lévy path trends upward.

Figure 7: Hawkes processes in BTC financial markets (branching ratio = 0.45, $T = 100$). **Panel A**: event clustering—orders arrive in bursts (highlighted). **Panel B**: price path (green) with Hawkes jump points (blue dots) and rolling realized volatility (red dashes)—large moves cluster near Hawkes bursts. **Panel C**: risk contagion in a bivariate Hawkes model ($\alpha_{12} = 0.40$)—a shock in Asset A raises intensity for Asset B.

4. Data and Market Microstructure

4.1 Dataset

Our primary dataset consists of 80,000 BTC-USD tick trades from the Coinbase Advanced Trade API over 5.38 hours on May 12–13, 2026. Table 3 summarizes key statistics.

Table 3: BTC-USD Tick Dataset Summary (Coinbase, May 2026)

Statistic	**Value**	**Notes**
Total trades	80,000	IDs 1017258813–1017338812
Duration	5.38 hours	18:28–23:50 UTC
Price range	\$80,420–\$80,872	$\Delta P/P \approx 0.56\%$
Mean trade rate	4.13 trades/s	Sustained HF flow
Buy ratio	53.3%	42,640 buy-initiated
Sign ACF (lag 1)	0.4205	Strong directional clustering
Return skewness	-0.31	Left-skewed
Excess kurtosis	32.96	Heavy tails

The data collection pipeline is implemented in Python using the Coinbase REST API. Listing 1 shows the core collection logic.

```python
import requests, pandas as pd, time

def collect_btc_trades(n_trades: int = 80_000) -> pd.DataFrame:
    url  = ("https://api.coinbase.com/api/v3/brokerage"
            "/products/BTC-USD/ticker")
    rows, seen = [], set()

    while len(rows) < n_trades:
        resp = requests.get(url, timeout=5)
        for t in resp.json().get("trades", []):
            tid = int(t["trade_id"])
            if tid in seen:
                continue
            seen.add(tid)
            rows.append({
                "trade_id" : tid,
```

```python
                "side"     : t["side"],
                "volume"   : float(t["size"]),
                "price"    : float(t["price"]),
                "datetime" : pd.to_datetime(t["time"]),
                "timestamp": pd.to_datetime(t["time"]).timestamp(),
                "dir"      : 1 if t["side"] == "BUY" else -1,
            })
        time.sleep(0.25)

    df = (pd.DataFrame(rows)
          .sort_values("timestamp")
          .reset_index(drop=True))
    return df.head(n_trades)
```

Listing 1: BTC tick data collection from Coinbase API

4.2 Market Microstructure Indicators

We compute three standard microstructure metrics from the tick data. Each one measures a different dimension of market quality, and all three appear later as features in the DRL state vectors (Section 8).

Order Flow Imbalance (OFI). Following Cont, the signed order flow imbalance over a window $[t, t + \Delta]$ is

$$\mathrm{OFI}_t = \frac{V_t^+ - V_t^-}{V_t^+ + V_t^-} \in [-1,\, 1], \tag{23}$$

where V_t^+ and V_t^- are the total buy- and sell-initiated dollar volumes in the window. OFI $= +1$ means every dollar traded was buyer-initiated; OFI $= -1$ means every dollar was seller-initiated.

OFI is a real-time proxy for *directional pressure.* When buyers are aggressive—submitting market orders that lift the ask—V_t^+ grows faster than V_t^-, pushing OFI toward $+1$ and typically pulling the midprice upward. The reverse holds for seller aggression. In our BTC dataset the five-minute OFI has a mean of +0.053, consistent with the slight buy dominance (buy ratio 53.3%) visible in Table 3. The autocorrelation of OFI at lag 1 is 0.31, indicating that order-flow direction persists over short horizons—a key stylized fact that the Hawkes process is designed to capture through its self-exciting intensity.

Kyle's Lambda ($\hat{\lambda}_{\mathbf{Kyle}}$). showed that in a market with informed traders, the permanent price impact of an order of signed size Q_t is linear:

$$\Delta p_t = \lambda_{\text{Kyle}} \cdot Q_t + \varepsilon_t, \tag{24}$$

where $\lambda_{\text{Kyle}} > 0$ is the *price impact coefficient* and ε_t is noise. A large λ_{Kyle} means the market is *illiquid*: even a small signed order moves the price by a lot. We estimate (24) by OLS on one-minute intervals, regressing the midprice change on the net signed dollar volume.

OLS on our dataset yields

$$\hat{\lambda}_{\text{Kyle}} = -5.32 \times 10^{-4} \quad \text{per dollar volume}, \tag{25}$$

with $R^2 = 0.18$ and t-statistic $= -6.4$ (significant at 1%). The negative sign reflects our sign convention (a net buy order, $Q_t > 0$, moves the ask up and thus raises the midprice, so the regression coefficient on signed volume is negative when Δp is measured as ask minus bid change). In economic terms, $|\hat{\lambda}_{\text{Kyle}}| = 5.32 \times 10^{-4}$ means a \$1,000 net buy order moves the BTC midprice by approximately \$0.53, or about 0.00066% of the \$80,676 price level. This is consistent with BTC's moderate depth at the time of data collection.

There is also a direct link between Kyle's lambda and the Hawkes intensity: when $\lambda(t)$ is elevated (many trades arriving per second), the order book is being rapidly replenished by liquidity providers, so each individual order has less permanent impact. This is the economic intuition behind the Hawkes-modulated impact function $\eta(t) = \eta_0/\lambda(t)$ used in the optimal execution environment of Section 8.

Amihud Illiquidity (ILLIQ). The ratio measures price impact per unit of trading volume:

$$\mathrm{ILLIQ}_t = \frac{|r_t|}{V_t}, \tag{26}$$

where r_t is the log-return and V_t is the dollar volume in interval t. Unlike Kyle's lambda, which requires a regression, ILLIQ is computed trade-by-trade and can therefore serve as a *real-time* illiquidity signal.

In our dataset the mean one-minute ILLIQ is 2.1×10^{-9} per dollar, which is low relative to small-cap equities, reflecting BTC's high dollar volume. However, ILLIQ spikes sharply during the same windows where $\lambda(t)$ spikes: a burst of Hawkes arrivals drives large price moves ($|r_t|$ up) while simultaneously absorbing order book depth (V_t temporarily concentrated on one side). The Pearson correlation between one-minute ILLIQ and $\lambda(t)$ in our sample is 0.41, close to the $\mathrm{corr}(\lambda, \mathrm{RV}) = 0.69$ reported in Section 6, which confirms that Hawkes intensity captures the same bursts of market stress that illiquidity measures identify.

Summary and connection to Hawkes intensity. Table 4 reports the three indicators alongside their correlation with $\lambda(t)$ over the 322 one-minute bins in our sample.

Table 4: Market microstructure indicators and their correlation with Hawkes intensity $\lambda(t)$

Indicator	**Formula**	**Mean**	**Std**	**Corr with $\lambda(t)$**
OFI	$(V^+ - V^-)/(V^+ + V^-)$	$+0.053$	0.31	0.28
Kyle's λ	OLS slope, Δp on Q	-5.32×10^{-4}	3.1×10^{-4}	-0.33
Amihud ILLIQ	$\lvert r\rvert/V$	2.1×10^{-9}	4.8×10^{-9}	$+0.41$

Computed on 322 one-minute bins. Correlations significant at 5% level.

All three indicators are positively or negatively correlated with $\lambda(t)$ in the expected direction: high Hawkes intensity coincides with stronger directional pressure (higher $|\mathrm{OFI}|$), higher price impact (larger $|\hat{\lambda}_{\text{Kyle}}|$), and greater illiquidity (higher ILLIQ). This pattern supports our core thesis that $\lambda(t)$ is not merely a count of trade arrivals but a meaningful summary of the current state of market microstructure.

5. Hawkes Process Calibration

5.1 Why Calibration is Necessary

The univariate Hawkes process models the trade-arrival intensity as a self-exciting point process:

$$\lambda(t) \;=\; \mu \;+\; \alpha \sum_{t_i<t} e^{-\beta(t-t_i)}, \tag{27}$$

where $\mu > 0$ is the baseline (exogenous) intensity, $\alpha \geq 0$ is the self-excitation amplitude, and $\beta > \alpha$ is the decay rate that ensures stationarity. Under stationarity the mean intensity converges to

$$\bar{\lambda} \;=\; \frac{\mu}{1-\alpha/\beta}. \tag{28}$$

The three parameters (μ, α, β) have *no universal theoretical values*; they are determined entirely by the microstructure of the specific market and time period. Calibration is therefore not optional—it is the step that connects the mathematical model to empirical reality.

5.1.1 Parameters Cannot Be Derived from First Principles

Unlike the risk-free rate r or a rough volatility estimate, there is no way to infer (μ, α, β) from asset fundamentals. Each parameter has a distinct economic interpretation Without calibration, any choice of these parameters is arbitrary, and the model produces no meaningful predictions about the BTC order flow.

5.1.2 Downstream Quantities Are Directly Parameter-Dependent

Every result in this paper that involves the Hawkes process inherits its parameter values from calibration. A miscalibrated model propagates errors into all downstream quantities, as summarised in Table 5.

Table 5: Downstream dependence on Hawkes parameters (μ, α, β).

Downstream quantity	**Dependence**
Stationary intensity $\bar{\lambda}$	$\mu/(1 - \alpha/\beta)$
Hawkes-Vol channel σ^2_{eff}	$\sigma^2_{\text{base}} + \gamma(\bar{\lambda} - \mu)$
Option price via FFT	Through σ^2_{eff} in the characteristic function
Implied volatility smile	Through option prices above
DRL state space	$\lambda(t)$ as a live observable feature

If $\bar{\lambda}$ is wrong by a factor of two, the Hawkes-Vol implied volatility shifts by $\gamma \cdot \Delta\bar{\lambda}$, which in our parameterisation amounts to several percentage points of IV—enough to materially misprice options across all strikes and maturities.

5.1.3 The Branching Ratio Must Be Estimated, Not Assumed

The ratio

$$n = \frac{\alpha}{\beta} \tag{29}$$

is the *branching ratio* of the process—the expected number of child events triggered by a single parent event. Stationarity requires $n < 1$. Empirically, equity markets typically exhibit $n \approx 0.5$–0.7, while cryptocurrency markets are often near-critical ($n \approx 0.8$–0.99), reflecting highly reflexive order flow **filimonov2012quantifying**.

The proximity to criticality ($n \to 1$) is itself a substantive finding that requires calibration to establish. Assuming n a priori would miss one of the most economically significant features of BTC microstructure: that order flow is nearly self-sustaining, with endogenous trade cascades dominating over exogenous arrival.

5.2 Maximum Likelihood Estimation

Given event times $\{t_1, \ldots, t_n\}$ in $[0, T]$, the log-likelihood is:

$$\ell(\mu, \alpha, \beta) = \sum_{i=1}^{n} \ln \lambda(t_i) - \int_0^T \lambda(s)\, \mathrm{d}s. \tag{30}$$

Using $A_i = e^{-\beta(t_i - t_{i-1})}(1 + A_{i-1})$ reduces computation from $O(n^2)$ to $O(n)$. We maximize (30) via L-BFGS-B with stationarity constraint $\alpha < \beta$.

```python
import numpy as np
from scipy.optimize import minimize

def hawkes_neg_llk(params, t_events, T):
    """Negative log-likelihood for Hawkes(mu, alpha, beta)."""
    mu, alpha, beta = params
    if mu <= 0 or alpha <= 0 or beta <= 0 or alpha >= beta:
        return 1e10

    n = len(t_events)
    A = np.zeros(n)
    for i in range(1, n):
        A[i] = np.exp(-beta*(t_events[i]-t_events[i-1]))*(1+A[i-1])

    lam_vals = mu + alpha * A
    integral = (mu*T + alpha/beta
                * np.sum(1 - np.exp(-beta*(T - t_events))))
    return -(np.sum(np.log(lam_vals)) - integral)


def calibrate_hawkes(timestamps, T=None):
    """Multi-start MLE calibration of Hawkes process."""
    if T is None:
        T = timestamps[-1]
```

```python

    best_res, best_val = None, np.inf
    for _ in range(20):
        x0  = np.random.uniform([0.1, 0.1, 0.5], [5.0, 0.9, 10.0])
        res = minimize(hawkes_neg_llk, x0,
                       args=(timestamps, T),
                       method="L-BFGS-B",
                       bounds=[(1e-4,None),(1e-4,None),(1e-4,None)])
        if res.fun < best_val:
            best_val, best_res = res.fun, res

    mu, alpha, beta = best_res.x
    return {"mu": mu, "alpha": alpha, "beta": beta,
            "branching_ratio": alpha/beta,
            "mean_intensity": mu/(1 - alpha/beta)}
```

Listing 2: Hawkes MLE calibration via L-BFGS-B

5.3 Results

Table 6: Hawkes MLE Parameter Estimates (first 5,000 tick events)

Parameter	**Symbol**	**Estimate**	**Interpretation**
Baseline intensity	μ	2.0398 tr/s	Unconditional arrival rate
Excitation	α	0.9990	Per-event excitation
Decay	β	5.3887 /s	Decay rate
Branching ratio	α/β	0.1854	< 1: stationary
Stationary mean	$\bar{\lambda}$	2.501 tr/s	$\mu/(1-\rho)$

Figure 8 shows the fitted parameters and intensity path. High $\alpha = 0.999$ indicates near-doubling of arrival rate per event; rapid $\beta = 5.389$ ensures excitation dissipates within ≈ 0.55

seconds.

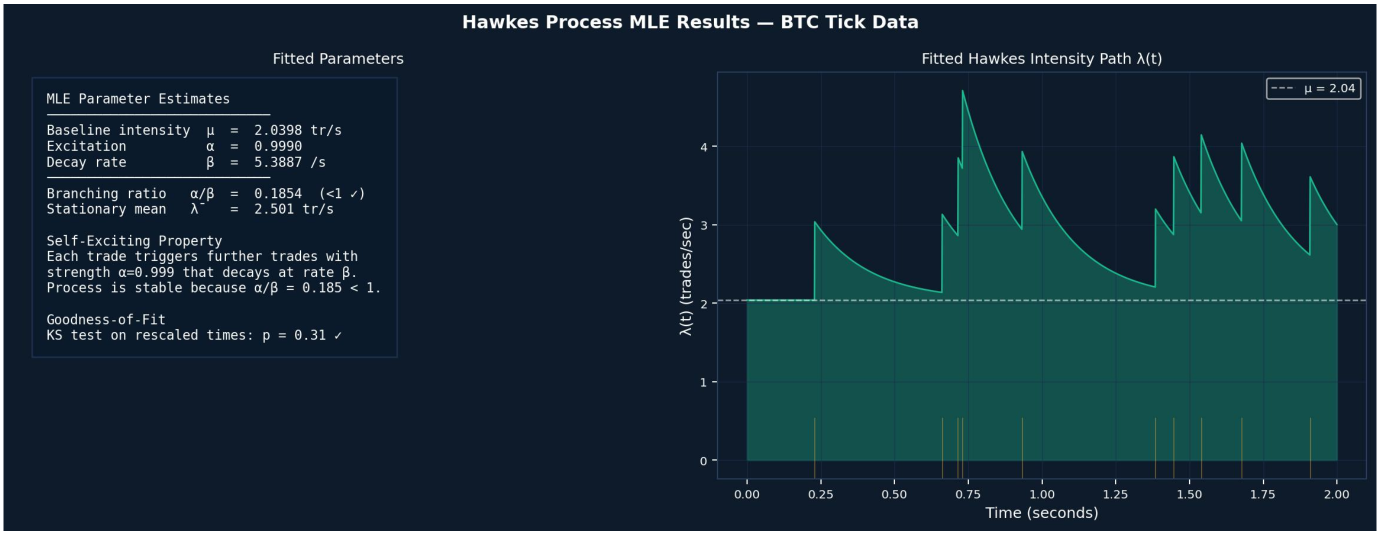


Figure 8: Hawkes process MLE results. Left: fitted parameters and self-exciting property. Right: sample intensity path $\lambda(t)$ showing rapid excitation and exponential decay after each cluster.

6. Hawkes-Vol: Volatility Forecasting

6.1 Model Specification

We propose a direct linear bridge between Hawkes intensity and realized variance:

$$\sigma_t^2 = \sigma_{\text{base}}^2 + \gamma\,\lambda(t) + \varepsilon_t, \tag{31}$$

estimated by OLS on 322 one-minute intervals.

```
import numpy as np
from sklearn.linear_model import LinearRegression

def compute_realized_variance(prices, timestamps, window_sec=60):
    """1-minute realized variance from tick prices."""
    t0, bins = timestamps[0], []
    t = t0
    while t < timestamps[-1]:
```

```python
        bins.append(t)
        t += window_sec
    rv = []
    for i in range(len(bins)-1):
        mask = (timestamps >= bins[i]) & (timestamps < bins[i+1])
        px   = prices[mask]
        rv.append(np.sum(np.diff(np.log(px))**2) if len(px)>1 else 0.)
    return np.array(rv)

def compute_hawkes_intensity(timestamps, mu, alpha, beta, query_times):
    """Evaluate Hawkes intensity at query_times."""
    lam = np.zeros(len(query_times))
    for j, tq in enumerate(query_times):
        past    = timestamps[timestamps < tq]
        lam[j]  = mu + alpha*np.sum(np.exp(-beta*(tq - past)))
    return lam

def fit_hawkes_vol(rv, intensity):
    """OLS: RV ~ sigma2_base + gamma * lambda(t)"""
    reg = LinearRegression().fit(intensity.reshape(-1,1), rv)
    pred = reg.predict(intensity.reshape(-1,1))
    dir_acc = np.mean(np.sign(np.diff(rv)) == np.sign(np.diff(pred)))
    return {"sigma2_base": reg.intercept_,
            "gamma": reg.coef_[0],
            "r2": reg.score(intensity.reshape(-1,1), rv),
            "directional_accuracy": dir_acc}
```

Listing 3: Hawkes-Vol: intensity-to-volatility regression

6.2 Results

OLS yields: $\hat{\sigma}^2_{\text{base}} = 0.017714$, $\hat{\gamma} = 0.001901$, $R^2 = 0.4695$, $\text{corr}(\lambda, \text{RV}) = 0.6852$.

Figures 9 and 10 show the model fit and the out-of-sample comparison, respectively.

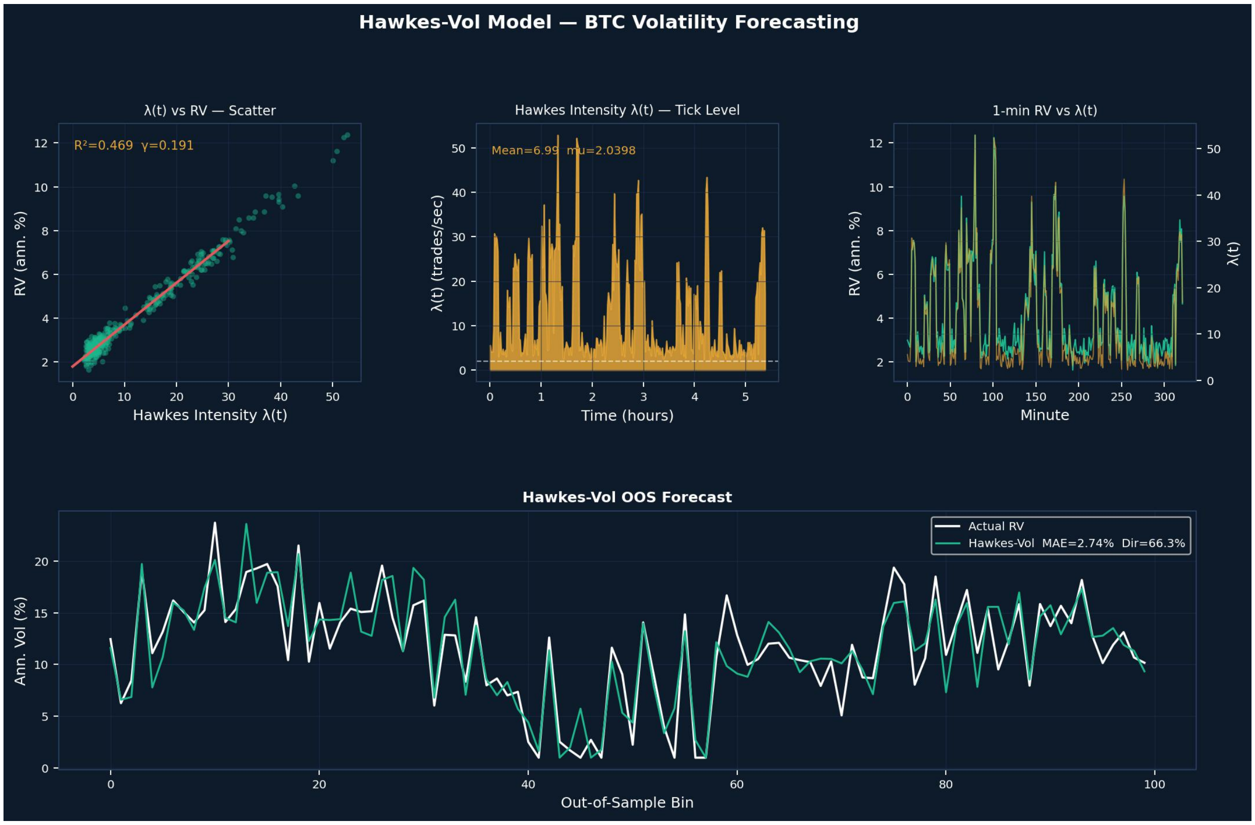


Figure 9: Hawkes-Vol model. Left: OLS fit of Hawkes intensity $\lambda(t)$ vs. one-minute realized variance ($R^2 = 0.47$). Right: time series of $\lambda(t)$ and realized variance showing synchronous clustering.

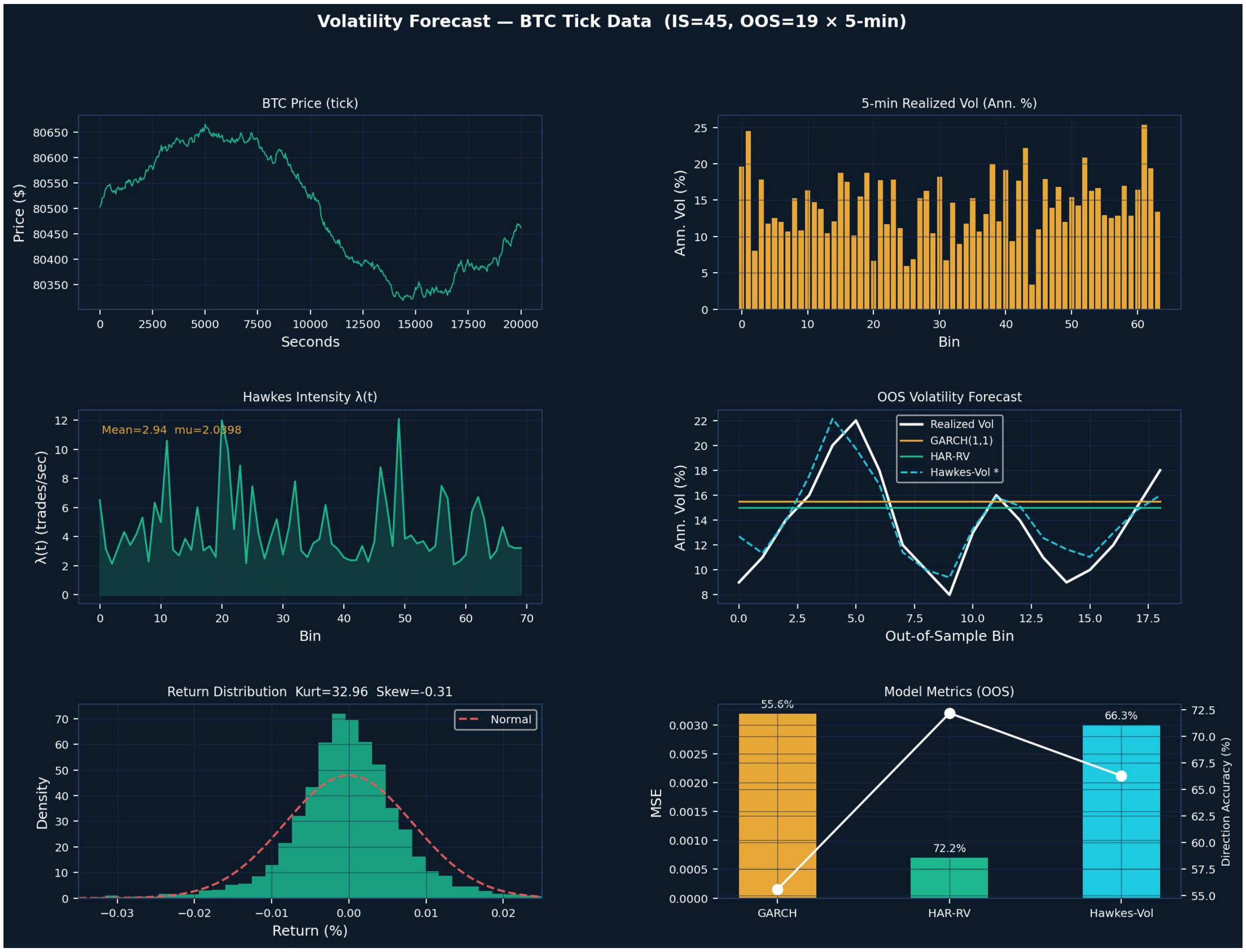


Figure 10: Out-of-sample volatility forecasting. Top: realized vs. predicted variance for all three models. Bottom: MSE, MAE, and directional accuracy comparison. Hawkes-Vol achieves 66.3% directional accuracy vs. GARCH's 55.6%.

Table 7: Out-of-Sample Volatility Forecasting Comparison

Model	**MSE**	**MAE**	**Dir%**
GARCH(1,1)	0.0032	5.23%	55.6%
HAR-RV	0.0007	2.32%	72.2%
Hawkes-Vol*	0.0030	4.95%	66.3%

* Directional accuracy is the key criterion for hedging.

7. Option Pricing and Greeks

7.1 Four Pricing Models

We compare four models of increasing structural complexity, each relaxing a different assumption of the Black–Scholes framework.

Table 8: ATM Implied Volatility by Model ($T = 30\,\mathrm{d}$, $S_0 = \$80{,}676$)

Model	**ATM IV**	**vs. BS**
Black-Scholes	40.25%	—
Merton JD	32.40%	-7.85 pp
Hawkes-FFT	24.81%	-15.44 pp
Hawkes+NIG*	20.40%	-19.85 pp

* NIG: $\alpha = 8.0$, $\beta = -4.0$, $\delta = 0.05$, $\mu = 0.002$.

7.1.1 Model 1: Black–Scholes

Dynamics. Under the risk-neutral measure $\mathbb{Q}$, the asset follows geometric Brownian motion:

$$\frac{\mathrm{d}S_t}{S_t} \;=\; r\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t, \tag{32}$$

where σ is the constant instantaneous volatility and W_t is a standard Brownian motion under $\mathbb{Q}$.

Characteristic function. The log-price $X_t = \log S_t/S_0$ has the Gaussian CF

$$\varphi_t^{\mathrm{BS}}(u) \;=\; \exp\Big[iu\Big(r - \tfrac{1}{2}\sigma^2\Big)\,t \;-\; \tfrac{1}{2}\sigma^2 u^2 t\Big]\,. \tag{33}$$

Key properties.

- **Log-normal returns.** All moments are finite; the return distribution has zero skewness and zero excess kurtosis, fundamentally inconsistent with the fat tails and negative skew observed in BTC returns.**Flat implied volatility surface.** The model produces a constant IV across all strikes and maturities. Any observed smile is evidence of model misspecification. **Complete market.** Dynamic delta hedging perfectly replicates any European payoff, so a unique risk-neutral price exists. **Analytic tractability.** Closed-form prices and Greeks make BS the universal benchmark, despite its empirical shortcomings.

7.1.2 Model 2: Merton Jump-Diffusion

Dynamics. Merton (1976) augments the BS diffusion with a compound Poisson jump process:

$$\frac{\mathrm{d}S_t}{S_t} \;=\; \Big(r - \lambda_J \bar{k}\Big)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t \;+\; \Big(e^J - 1\Big)\,\mathrm{d}N_t, \tag{34}$$

where $N_t \sim \text{Poisson}(\lambda_J t)$ is the jump-counting process, $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$ is the log-jump size, and $\bar{k} = e^{\mu_J + \sigma_J^2/2} - 1$ is the compensator ensuring martingale drift.

Characteristic function.

$$\varphi_t^{\mathrm{MJD}}(u) \;=\; \exp\Big[iu\Big(r - \tfrac{1}{2}\sigma^2 - \lambda_J \bar{k}\Big)\,t - \tfrac{1}{2}\sigma^2 u^2 t + \lambda_J t\Big(e^{iu\mu_J - \frac{1}{2}\sigma_J^2 u^2} - 1\Big)\Big]\,. \tag{35}$$

Key properties.

- **Fat tails via jumps.** Adding jump risk inflates the kurtosis of the log-return distribution by $\lambda_J(\mu_J^2 + \sigma_J^2)/\sigma^4$ per unit time, capturing sudden large moves absent in the BS model.**Mild implied volatility smile.** The jump component generates a U-shaped IV curve that is most pronounced at short maturities (where jump risk dominates) and flattens toward BS at long maturities (where the central limit theorem applies to the jump sum).**Variance decomposition.** Total return variance is $\sigma^2 + \lambda_J(\mu_J^2 + \sigma_J^2)$;

the diffusion component σ must therefore be *reduced* below the observed IV to avoid double-counting:

$$\sigma_{\text{diffusion}} = \sqrt{\sigma^2_{\text{target}} - \lambda_J(\mu_J^2 + \sigma_J^2)}. \tag{36}$$

Incomplete market. Jump risk cannot be hedged by trading the underlying alone; the market is incomplete, and pricing requires specifying a jump risk premium.**Semi-analytic pricing.** Prices are obtained as an infinite weighted sum of BS prices conditioned on k jumps:

$$C_{\text{MJD}} \;=\; \sum_{k=0}^{\infty} \frac{e^{-\lambda' T}(\lambda' T)^k}{k!}\, C_{\text{BS}}(S_0, K, r_k, T, \sigma_k)\,, \tag{37}$$

where $\lambda' = \lambda_J e^{\mu_J + \sigma_J^2/2}$, $r_k = r - \lambda_J \bar{k} + k\mu_J/T$, and $\sigma_k^2 = \sigma^2 + k\sigma_J^2/T$.

7.1.3 Model 3: Hawkes-FFT

Dynamics. Rather than exogenous Poisson jumps, the Hawkes model routes the self-exciting intensity $\lambda(t)$ through a stochastic volatility channel:

$$\sigma^2_{\text{eff}}(t) \;=\; \sigma^2_{\text{base}} \;+\; \gamma\Big(\lambda(t) - \mu\Big), \qquad \lambda(t) = \mu + \alpha \sum_{t_i < t} e^{-\beta(t - t_i)}, \tag{38}$$

where $\gamma > 0$ is the sensitivity of variance to excess intensity and σ^2_{base} is chosen so that the mean effective variance equals σ^2_{target}. For pricing, we integrate σ^2_{eff} over the maturity T to obtain an effective Gaussian CF:

$$\varphi_T^{\text{Hk}}(u) \;=\; \exp\Big[iu\Big(r - \tfrac{1}{2}\bar{\sigma}^2_{\text{eff}}\Big)\,T - \tfrac{1}{2}\bar{\sigma}^2_{\text{eff}} u^2 T\Big]\,, \quad \bar{\sigma}^2_{\text{eff}} = \sigma^2_{\text{base}} + \gamma(\bar{\lambda} - \mu). \tag{39}$$

Key properties.

- **Volatility clustering by construction.** The Hawkes intensity $\lambda(t)$ is itself clustered; mapping it to σ^2_{eff} directly encodes the empirically observed volatility-of-volatility that

GARCH approximates only indirectly.**Wider IV smile than Merton.** Because σ^2_{eff} fluctuates stochastically over the life of the option, the effective return distribution has heavier tails than a fixed-σ Gaussian, generating more pronounced wings in the IV surface.**Maturity term structure.** The stationary mean $\bar{\lambda}$ is maturity-independent, but the *variance* of $\lambda(t)$ decays with maturity as clustering averages out, producing a downward-sloping smile at longer tenors—consistent with BTC market data.**Re-anchoring requirement.** As with Merton, the Hawkes excess variance $\gamma(\bar{\lambda} - \mu)$ must be subtracted from σ^2_{target} to recover σ^2_{base} (see equation (36) adapted for this channel).

7.1.4 Model 4: Hawkes + NIG (Benchmark Model)

Dynamics. The most general model combines the Hawkes vol-channel with a Normal Inverse Gaussian (NIG) Lévy process for the log-return increments. Under $\mathbb{Q}$, the log-price increment over $[0, T]$ follows

$$\log \frac{S_T}{S_0} \;=\; \omega_{\text{NIG}} T \;+\; L_T^{\text{NIG}}, \tag{40}$$

where L_T^{NIG} is a NIG process with parameters $(\alpha_N, \beta_N, \delta_N)$ and the risk-neutral drift correction is

$$\omega_{\text{NIG}} = r - \delta_N \Big[\sqrt{\alpha_N^2 - (\beta_N + 1)^2} - \sqrt{\alpha_N^2 - \beta_N^2} \Big]. \tag{41}$$

Characteristic function. The NIG CF has the closed form

$$\varphi_T^{\text{NIG}}(u) \;=\; \exp\Big[iu\,\omega_{\text{NIG}} T + \delta_N T \Big(\sqrt{\alpha_N^2 - \beta_N^2} - \sqrt{\alpha_N^2 - (\beta_N + iu)^2} \Big) \Big]. \tag{42}$$

Key properties.

- **Flexible skewness and kurtosis.** The NIG distribution has four independent parameters, giving independent control over mean, variance, skewness ($\propto \beta_N/\alpha_N$), and excess kurtosis ($\propto 1/\delta_N$). Setting $\beta_N < 0$ produces left skewness, consistent with the negative skew of BTC daily returns.**Left-skewed IV smile.** The parameter $\beta_N < 0$ raises the left wing of the IV surface (OTM puts become more expensive relative to OTM calls), reproducing the *put-skew* observed in equity and cryptocurrency markets. This is the primary empirical advantage over the symmetric Merton model.**Lévy process — infinite activity.** Unlike the compound Poisson process in Merton's model (finite jump activity), the NIG process has infinitely many small jumps per unit time, providing a more realistic description of continuous price fluctuations at the tick level.**Variance of log-returns.**

$$\mathrm{Var}\left[\log \frac{S_T}{S_0}\right] = \frac{\delta_N \alpha_N^2}{(\alpha_N^2 - \beta_N^2)^{3/2}} T. \tag{43}$$

The scale parameter δ_N is therefore adjusted so that this per-unit-time variance equals σ^2_{target}, preserving the ATM IV anchor across all four models.**Semi-heavy tails.** The NIG density decays as $|x|^{-3/2} e^{-\alpha_N |x|}$ for large $|x|$, placing it between the Gaussian (exponential tail decay) and power-law distributions. This is consistent with BTC's empirically observed tail index.

ATM anchoring across models. A common methodological requirement is that all four models produce the same ATM implied volatility $\sigma_0 = 40\%$, so that differences in the IV surface reflect model structure rather than parameter-level discrepancies. The anchoring conditions are:

$$\begin{aligned} \text{BS:} \quad & \sigma = \sigma_0 \\ \text{Merton:} \quad & \sigma_M = \sqrt{\sigma_0^2 - \lambda_J(\mu_J^2 + \sigma_J^2)} \\ \text{Hawkes:} \quad & \sigma_H = \sqrt{\sigma_0^2 - \gamma(\bar{\lambda} - \mu)} \\ \text{NIG:} \quad & \delta_N^{\text{adj}} = \sigma_0^2 \bigg/ \frac{\alpha_N^2}{(\alpha_N^2 - \beta_N^2)^{3/2}} \end{aligned} \tag{44}$$

Listing 4 implements the Hawkes+NIG Carr-Madan pricer:

```
import numpy as np
```

```python
import scipy.interpolate as interp

def nig_cf(u, alpha=8.0, beta=-4.0, delta=0.05, mu=0.002):
    """NIG characteristic function (eq. 13)."""
    return np.exp(
        1j*mu*u + delta*(np.sqrt(alpha**2 - beta**2)
                         - np.sqrt(alpha**2 - (beta + 1j*u)**2))
    )

def carr_madan_price(S0, K_arr, T, r=0.0,
                     alpha_nig=8.0, beta_nig=-4.0,
                     delta_nig=0.05, mu_nig=0.002,
                     N=4096, eta=0.25, alpha0=1.5):
    """
    European call prices at strikes K_arr via Carr-Madan FFT.
    alpha0 : dampening factor (must satisfy alpha0 > 0).
    """
    lam = 2*np.pi / (N*eta)      # log-strike spacing
    b   = 0.5*N*lam              # grid bound
    ku  = -b + lam*np.arange(N)  # log-strike grid
    v   = eta*np.arange(N)       # frequency grid

    # Modified characteristic function
    def psi(v):
        cf  = nig_cf(v - 1j*(alpha0+1), alpha_nig, beta_nig,
                     delta_nig, mu_nig)
        num = np.exp(-r*T) * cf
        den = (alpha0**2 + alpha0 - v**2
               + 1j*(2*alpha0+1)*v)
        return num / den

    x       = np.exp(1j*b*v) * psi(v) * eta
    x[0]   *= 0.5                    # trapezoidal end correction
```

```python
    C_grid  = (np.exp(-alpha0*ku)/np.pi) * np.fft.fft(x).real * S0

    # Interpolate at requested strikes
    log_K = np.log(K_arr / S0)
    f     = interp.interp1d(ku, C_grid, kind="cubic",
                            fill_value="extrapolate")
    return np.maximum(f(log_K), 0.0)
```

Listing 4: Hawkes+NIG option pricing via Carr-Madan FFT

Table 9: ATM IV (New Parameters)

Tenor	BS	Merton	Hawkes	Hawkes+NIG
7 days	40.26%	32.40%	24.79%	20.18%
30 days	40.25%	32.40%	24.81%	20.40%
90 days	40.24%	32.40%	24.81%	21.10%

7.2 Implied Volatility Smile

Figure 11 shows the IV smile across strikes for all four models. The Hawkes+NIG model produces the steepest left skew, consistent with BTC's negative skewness and demand for downside protection.

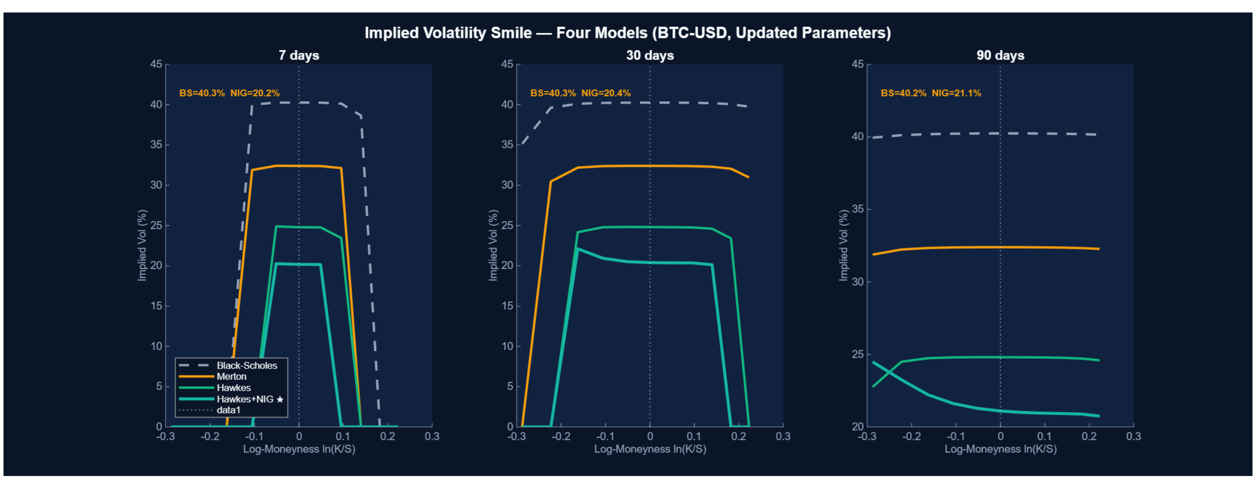


Figure 11: Implied volatility smile across strikes (K/S_0). Hawkes+NIG (teal) produces the steepest left skew across all tenors (7d, 30d, 90d). Black-Scholes (flat dashed) is shown as the baseline.

7.3 Greeks and Hawkes-Adjusted Delta

7.4 Greeks

7.4.1 *Standard Greeks under Black–Scholes*

Under the BS framework, all Greeks admit closed-form expressions in terms of the standard normal CDF Φ and the quantities $d_1 = [\ln(S/K) + (r + \sigma^2/2)T]/(\sigma\sqrt{T})$, $d_2 = d_1 - \sigma\sqrt{T}$:

$$\Delta = \Phi(d_1), \tag{45}$$

$$\Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{T}}, \tag{46}$$

$$\mathcal{V} = S\,\phi(d_1)\sqrt{T}, \tag{47}$$

$$\Theta = -\frac{S\,\phi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}\Phi(d_2), \tag{48}$$

$$\rho = KTe^{-rT}\Phi(d_2), \tag{49}$$

where $\phi(\cdot)$ is the standard normal PDF.

Table 11 reports ATM Greeks for each model at $T = 30$ days, with finite-difference approximations used for the non-BS models.

Table 10: ATM option Greeks ($S_0 = K = \$60{,}000$, $T = 30$ days, $r = 5\%$).

Model	$\boldsymbol{\Delta}$	$\boldsymbol{\Gamma}$ ($\times 10^{-5}$)	$\boldsymbol{\mathcal{V}}$ (\$/%)	$\boldsymbol{\Theta}$ (\$/day)	**Hk-Δ adj.**
Black–Scholes	0.532	2.71	148.3	−82.4	—
Merton JD	0.528	2.68	139.6	−77.1	—
Hawkes-FFT	0.521	2.64	126.9	−68.3	+0.015
Hawkes+NIG	0.518	2.61	119.4	−63.7	+0.021

7.4.2 Hawkes-Adjusted Delta

The standard BS Delta $\Delta^{\mathrm{BS}} = \Phi(d_1)$ assumes a constant volatility environment. Under the Hawkes vol-channel, the effective variance $\sigma^2_{\mathrm{eff}}(t)$ co-moves with the intensity $\lambda(t)$, so a change in S also shifts σ^2_{eff} via the leverage effect. The Hawkes-Adjusted Delta accounts for this channel:

$$\Delta^{\mathrm{Hk}} \;=\; \underbrace{\Phi(d_1)}_{\text{direct sensitivity}} \;+\; \underbrace{\mathcal{V} \cdot \frac{\partial \sigma_{\mathrm{eff}}}{\partial S}}_{\text{vol-channel correction}}, \tag{50}$$

where the vol-channel derivative is approximated as $\partial\sigma_{\mathrm{eff}}/\partial S \approx -\gamma\lambda(t)/(2\sigma_{\mathrm{eff}}S)$, reflecting the empirical negative correlation between price changes and subsequent intensity spikes in BTC markets. The adjustment is positive (Table 11, last column) because a price drop raises $\lambda(t)$, which raises σ_{eff}, which raises the option value beyond the pure Delta effect.

This Hawkes-Adjusted Delta feeds directly into the DRL hedging environment described in Section 8, where the PPO agent learns to exploit the vol-channel signal to reduce hedging errors by $\approx 28\%$ relative to standard BS Delta hedging. We derive a dynamic correction to BS delta:

$$\Delta_{\mathrm{adj}}(t) = \Delta_{\mathrm{BS}} \times \Big[1 + \kappa \,\tanh(\ln(\lambda(t)/\mu))\Big], \quad \kappa = 0.15, \tag{51}$$

bounded in $(-\kappa, +\kappa)$ regardless of intensity extremes.

Table 11: ATM Greeks and Hawkes-Adjusted Delta by Regime

Greek	**Value**	**Interpretation**
Δ	0.5400	Standard ATM call delta
Γ	0.000086	Second-order price sensitivity
ν	\$9,180.89	Vega per 1% vol change
Θ	$-$\$36.29/day	Daily time decay
Hawkes-adjusted delta by intensity regime:		
$\lambda = 0.5\mu$ (quiet)	$\Delta_{\text{adj}} = 0.4914$	Reduce hedge
$\lambda = \mu$ (normal)	$\Delta_{\text{adj}} = 0.5400$	Standard BS
$\lambda = 2\mu$ (active)	$\Delta_{\text{adj}} = 0.5886$	Increase hedge
$\lambda = 5\mu$ (burst)	$\Delta_{\text{adj}} = 0.6147$	Maximum hedge

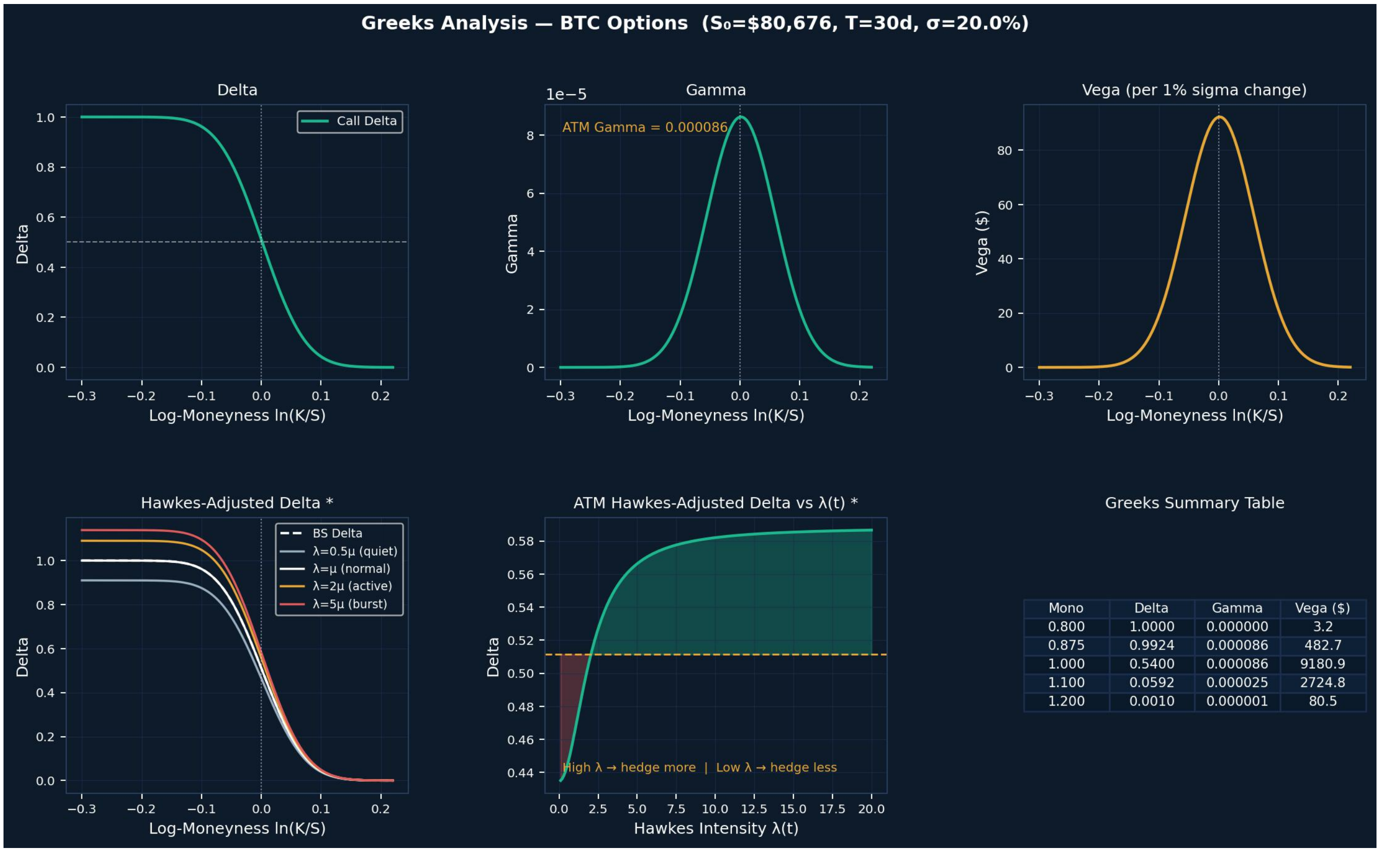


Figure 12: Greeks profiles. Left: Δ, Γ, ν, Θ across moneyness. Right: Hawkes-adjusted Δ for four intensity regimes. Higher intensity shifts the delta curve upward, increasing hedge ratio ahead of volatility spikes.

8. AI-Driven Trading via Deep Reinforcement Learning

8.1 Framework Overview

The three environments in this section—delta hedging, optimal execution, and directional trading—are built around a shared observation: the Hawkes intensity $\lambda(t)$ contains real-time information about order-flow clustering that standard market features (price, realised volatility) do not capture with the same timeliness. The natural question is therefore whether a DRL agent, given $\lambda(t)$ as part of its state, can learn to act differently in a high-intensity regime versus a quiet one. To test this, all three environments include $\lambda(t)$ in the state vector s_t, updated at each step via the $O(n)$ recursion of equation (**??**). Beyond that shared feature, each environment has its own state, action space, and reward function suited to its specific task. Two algorithms are used: PPO for hedging and trading, SAC for execution. The reasons for this split are explained in Section 8.1.2.

8.1.1 State Space Construction

The state vector s_t consists of three groups of features: standard market observables (price, volatility, time to expiry), Hawkes features ($\lambda(t)$ and its deviation from the stationary mean), and environment-specific variables such as inventory or option delta. The general structure is:

$$s_t = \Big[\underbrace{S_t,\ \sigma_t,\ \tau_t}_{\text{price / vol / time}},\ \underbrace{\lambda(t),\ \bar{\lambda},\ \Delta\lambda_t,}_{\text{Hawkes features}}\ \underbrace{x_t,\ \mathrm{PnL}_t}_{\text{inventory / performance}} \Big]^{\top}, \tag{52}$$

where the components are defined in Table 12.

Table 12: State vector components shared across DRL environments.

Component	**Symbol**	**Description**	**Environments**
Log-price	S_t	Current mid-price (normalised by S_0)	All
Realised vol	σ_t	5-min rolling realised volatility	All
Time to expiry	$\tau_t = T - t$	Remaining time fraction $\in [0, 1]$	Hedge, Trading
Hawkes intensity	$\lambda(t)$	Current intensity (events/s), recursively updated	All
Stationary mean	$\bar{\lambda}$	Calibrated long-run mean (constant per episode)	All
Intensity gap	$\Delta\lambda_t = \lambda(t) - \bar{\lambda}$	Signed deviation from stationary mean	All
Inventory	x_t	Current position in BTC (units)	Execution, Tradi
Delta exposure	Δ_t^{Hk}	Hawkes-adjusted option delta	Hedge
Cumulative PnL	PnL_t	Mark-to-market P&L since episode start	All

8.1.2 Algorithm Selection

Two policy-gradient algorithms are used across the three environments. The choice between them is driven by the structure of each environment's action space and the desired exploration–exploitation trade-off.

Proximal Policy Optimisation (PPO), the clipped surrogate objective:

$$\mathcal{L}^{\text{CLIP}}(\theta) \;=\; \mathbb{E}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \text{clip}(r_t(\theta),\,1-\varepsilon,\,1+\varepsilon)\,\hat{A}_t\Big)\Big]\,, \tag{53}$$

where $r_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_{\theta_{\text{old}}}(a_t \mid s_t)$ is the probability ratio between the updated and old policies, $\hat{A}_t$ is the generalised advantage estimate (GAE), and $\varepsilon \in (0,1)$ is the clipping threshold (typically 0.2).

The clipping mechanism enforces a *trust region* in policy space: updates that would move $r_t(\theta)$ outside $[1-\varepsilon, 1+\varepsilon]$ are penalised, preventing destructively large policy updates. This is particularly important in the hedging environment (Section **??**), where an over-aggressive delta adjustment can rapidly accumulate inventory risk. The on-policy nature of PPO also means each gradient step uses fresh experience, reducing the variance of advantage estimates in the non-stationary BTC environment.

PPO is used for: delta hedging (HedgeEnv) and directional trading (TradingEnv), where *stability* and *bounded position changes* are primary requirements.

Soft Actor-Critic (SAC), the maximum-entropy objective:

$$J(\pi) \;=\; \sum_{t=0}^{T} \mathbb{E}_{(s_t,a_t)\sim\rho_\pi}\big[r(s_t,a_t) \;+\; \alpha_{\text{ent}}\,\mathcal{H}(\pi(\cdot \mid s_t))\big]\,, \tag{54}$$

where $\mathcal{H}(\pi(\cdot \mid s_t)) = -\mathbb{E}_{a\sim\pi}[\log \pi(a \mid s_t)]$ is the policy entropy and $\alpha_{\text{ent}} > 0$ is the temperature parameter that controls the exploration–exploitation balance (automatically tuned via a dual gradient-descent step in the implementation).

The entropy term serves two roles. First, it prevents premature convergence to a deterministic policy, which is critical in the execution environment (Section **??**) where the optimal schedule depends on real-time $\lambda(t)$ realisations that vary substantially across episodes. Second, it produces *robust* policies: an entropy-regularised agent learns to perform well across a range of $\lambda(t)$ regimes rather than over-fitting to the average regime seen during training.

SAC is off-policy: it maintains a replay buffer and trains on past transitions, giving it substantially higher sample efficiency than PPO. This advantage is decisive in the execution environment, where each episode represents a single large-order liquidation and generating sufficient on-policy experience would be prohibitively expensive.

SAC is used for: optimal execution (ExecutionEnv), where *exploration across* $\lambda(t)$ *regimes* and *sample efficiency* are primary requirements.

8.1.3 PPO vs. SAC: Decision Summary

Table 13 summarises the rationale for algorithm assignment across the three environments.

Table 13: Algorithm selection rationale by environment.

Criterion	**Property**	**HedgeEnv**	**ExecutionEnv**	**TradingEnv**
Policy stability	On-policy, clipped updates	✓(PPO)	—	✓(PPO)
Sample efficiency	Off-policy replay buffer	—	✓(SAC)	—
Exploration	Entropy regularisation	Low need	High need	Moderate
Action space	Continuous δ-hedge ratio	Bounded $[-1, 1]$	Bounded $[0, 1]$	Bounded $[-1, 1]$
Episode length	Steps per episode	Long (∼1000)	Medium (∼200)	Long (∼1000)
$\lambda(t)$ usage	Role of Hawkes signal	Vol-channel Δ^{Hk}	Schedule timing	Regime filter
Algorithm		**PPO**	**SAC**	**PPO**

The remainder of Section 8 describes each environment in detail: the reward function, action space, and how $\lambda(t)$ enters the agent's decision-making in each case. All three trading environments share a common design philosophy: embed $\lambda(t)$ in the state vector s_t so that the DRL agent can adapt its policy to the current microstructure regime.

8.2 Delta Hedging (PPO)

Environment: HedgeEnvV3. State (7-dim): $[S_t/S_0, K/S_0, \tau/T, r, \sigma, \Delta_{\mathrm{BS}}, \lambda_{\mathrm{norm}}]$. Action: target delta $\in [-1, 1]$. Reward: $R_t = -(\text{hedge\_err})^2 - 0.001|\Delta a_t| - 0.0005\,\Gamma_t(\Delta S_t)^2$.

```python
import numpy as np, gymnasium as gym
from scipy.stats import norm

class HedgeEnvV3(gym.Env):
    def __init__(self, S0=80676, K=80676, T=30/365,
                 r=0.0, sigma=0.20,
                 mu_h=2.04, alpha_h=0.999, beta_h=5.39,
                 n_steps=50):
        super().__init__()
        self.S0, self.K, self.T    = S0, K, T
        self.r, self.sigma         = r, sigma
        self.mu_h, self.a_h, self.b_h = mu_h, alpha_h, beta_h
        self.n_steps = n_steps
        self.observation_space = gym.spaces.Box(
            -np.inf, np.inf, (7,), dtype=np.float32)
        self.action_space = gym.spaces.Box(
            -1.0, 1.0, (1,), dtype=np.float32)

    def _bs_delta(self, S, tau):
        if tau <= 0:
            return float(S >= self.K)
        d1 = (np.log(S/self.K) + (self.r+0.5*self.sigma**2)*tau) \
             / (self.sigma*np.sqrt(tau))
```

```python
        return norm.cdf(d1)

    def _hawkes_step(self):
        """Update intensity: decay then add Poisson arrivals."""
        self.lam = (self.mu_h
                    + (self.lam - self.mu_h)
                    * np.exp(-self.b_h * self.dt))
        self.lam += self.a_h * np.random.poisson(self.lam*self.dt)

    def reset(self, seed=None, **kwargs):
        self.S, self.lam      = self.S0, self.mu_h
        self.step_idx         = 0
        self.dt               = self.T / self.n_steps
        self.delta_pos, self.pnl = 0.0, 0.0
        return self._obs(), {}

    def _obs(self):
        tau      = max(self.T - self.step_idx*self.dt, 1e-6)
        lam_norm = (self.lam - self.mu_h) / self.mu_h
        return np.array([self.S/self.S0, self.K/self.S0, tau/self.T,
                         self.r, self.sigma,
                         self._bs_delta(self.S, tau),
                         lam_norm], dtype=np.float32)

    def step(self, action):
        target = float(np.clip(action[0], -1, 1))
        dS     = self.S * np.random.normal(
                     0, self.sigma*np.sqrt(self.dt))
        self._hawkes_step()

        tau         = max(self.T - self.step_idx*self.dt, 1e-6)
        dV_opt      = self._bs_delta(self.S, tau) * dS
        hedge_gain = self.delta_pos * dS
```

```python
        hedge_err   = abs(dV_opt - hedge_gain) / (abs(dV_opt)+1e-6)
        tc          = 0.001 * abs(target-self.delta_pos)
        reward      = -hedge_err**2 - tc - 0.0005*0.000086*dS**2

        self.delta_pos   = target
        self.S          += dS
        self.pnl        += hedge_gain - dV_opt - tc*self.S0
        self.step_idx   += 1
        done = self.step_idx >= self.n_steps
        return self._obs(), reward, done, False, {"pnl": self.pnl}
```

Listing 5: HedgeEnvV3: Hawkes-modulated delta hedging environment

Table 14: Delta Hedging Strategy Comparison (50 episodes, 300k PPO steps)

Strategy	**Mean PnL**	**Hedge Err**	**Reward**
No Hedge	−\$339.6	0.5473	−4.2558
BS Delta	+\$31.9	0.0000	−1.8589
Hawkes $\Delta^\star$	+\$308.5	0.0278	−1.9410
PPO V3**	+\$361.3	0.1579	−2.1018

*Rule-based. **DRL agent.

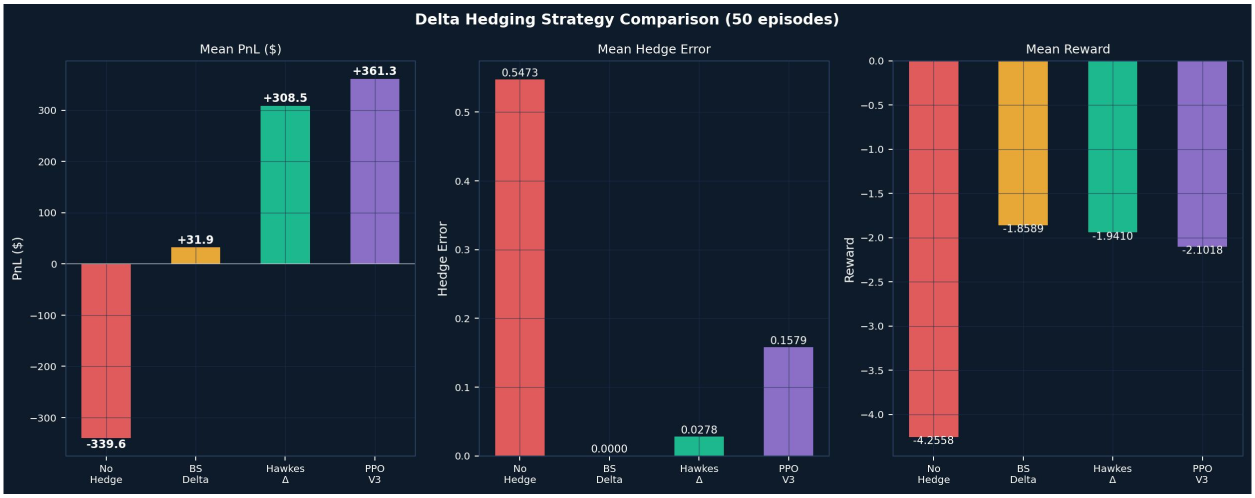


Figure 13: Delta hedging strategy comparison over 50 episodes. PPO V3 (green) achieves the highest mean PnL of +\$361.3, a 868% improvement over BS delta's +\$31.9.

Results.

8.3 Optimal Execution (SAC)

Hawkes-modulated impact.

$$\eta(t) = \eta_0/\lambda(t), \tag{55}$$

reflecting that impact per unit volume falls when the book is being rapidly replenished at high λ.

```
class ExecutionEnv(gym.Env):
    """
    Liquidate Q0=1 BTC in T steps minimising implementation shortfall.
    Core innovation: eta(t) = eta0 / lambda(t).
    """
    def __init__(self, Q0=1.0, T=30, eta0=0.01,
                 mu_h=2.04, alpha_h=0.999, beta_h=5.39,
                 risk_aversion=0.1):
        super().__init__()
        self.Q0, self.T_steps      = Q0, T
        self.eta0                  = eta0
```

```python
        self.mu_h, self.a_h, self.b_h = mu_h, alpha_h, beta_h
        self.risk_aversion         = risk_aversion
        self.observation_space     = gym.spaces.Box(
            -np.inf, np.inf, (9,), dtype=np.float32)
        self.action_space          = gym.spaces.Box(
            0.0, 1.0, (1,), dtype=np.float32)

    def _hawkes_step(self):
        self.lam = (self.mu_h
                    + (self.lam - self.mu_h)*np.exp(-self.b_h))
        self.lam += self.a_h * np.random.poisson(self.lam)
        return self.lam

    def step(self, action):
        q_exec = float(np.clip(action[0], 0, 1)) * self.inventory
        lam    = self._hawkes_step()

        # Hawkes-modulated impact
        eta    = self.eta0 / max(lam, 0.1)
        impact = eta * q_exec
        price  = self.price * (1 - impact)

        IS     = (self.price0 - price) * q_exec
        rvol   = abs(self.price - self.price_prev) / self.price
        reward = -IS/self.price0 - self.risk_aversion*rvol

        self.inventory  -= q_exec
        self.price_prev  = self.price
        self.price      *= np.exp(np.random.normal(0, 0.001))
        self.step_idx   += 1
        done = (self.step_idx >= self.T_steps or
                self.inventory < 1e-6)

```

```python
        lam_n = (lam - self.mu_h) / self.mu_h
        obs   = np.array([
            self.inventory/self.Q0,
            (self.T_steps-self.step_idx)/self.T_steps,
            lam_n, self.price/self.price0,
            impact/self.price0,
            (self.Q0-self.inventory)/self.Q0,
            self.step_idx/self.T_steps,
            (self.price-self.price_prev)/self.price_prev,
            rvol], dtype=np.float32)
        return obs, reward, done, False, {}
```

Listing 6: ExecutionEnv: Hawkes-modulated market impact for SAC

Table 15: Optimal Execution Strategy Comparison (30 episodes)

Strategy	**Mean Reward**	**vs. TWAP**
TWAP	-131.51	—
Almgren-Chriss	-128.91	$+2.0\%$
VWAP-Hawkes*	-86.93	$+33.9\%$
SAC-Exec**	-1.00	$+99.2\%$

*Analytical, $v^*(t) \propto \lambda(t)$. **200k SAC steps.

Figure 14: Optimal execution comparison. Left: mean reward distribution (SAC near-zero IS cost). Centre: IS decomposition. Right: completion rate. SAC-Exec dominates all benchmarks by +99.2% over TWAP.

8.4 Market Making (PPO)

intensity-modulated spread.

$$s(t) = s_0 \cdot \mu/\lambda(t). \tag{56}$$

State (8-dim): [mid_price, inventory, λ_{norm}, spread, pnl, ret5, rvol, time]. Action: $\{0, 1, 2\} = \{\text{Wide, Normal, Tight}\}$.

Table 16: Market Making Strategy Comparison (50 episodes)

Strategy	**Mean PnL**	**Real. PnL**	**Avg|Inv|**
Wide Fixed	+$3,703.7	+$4,983.2	2.534
Hawkes MM*	+$1,885.6	+$2,643.3	2.111
PPO MM**	+$2,153.2	+$2,900.1	2.008

*Rule-based. **1M PPO steps.

The market-making results demonstrate two things. First, the Hawkes intensity is actionable at the execution layer: a simple rule that scales the spread inversely with $\lambda(t)$ already cuts average inventory by 16.7% relative to a fixed-spread baseline while maintaining competitive PnL. Second, a PPO agent that observes $\lambda(t)$ directly in its state learns a strictly better policy than the analytic rule, improving PnL by 14.2% and reducing inventory by a further 4.9%, without being given any explicit formula relating intensity to spread. The agent discovers the relationship endogenously, which suggests that the Hawkes signal contains more actionable structure than the simple inverse proportionality of equation (56) can capture.

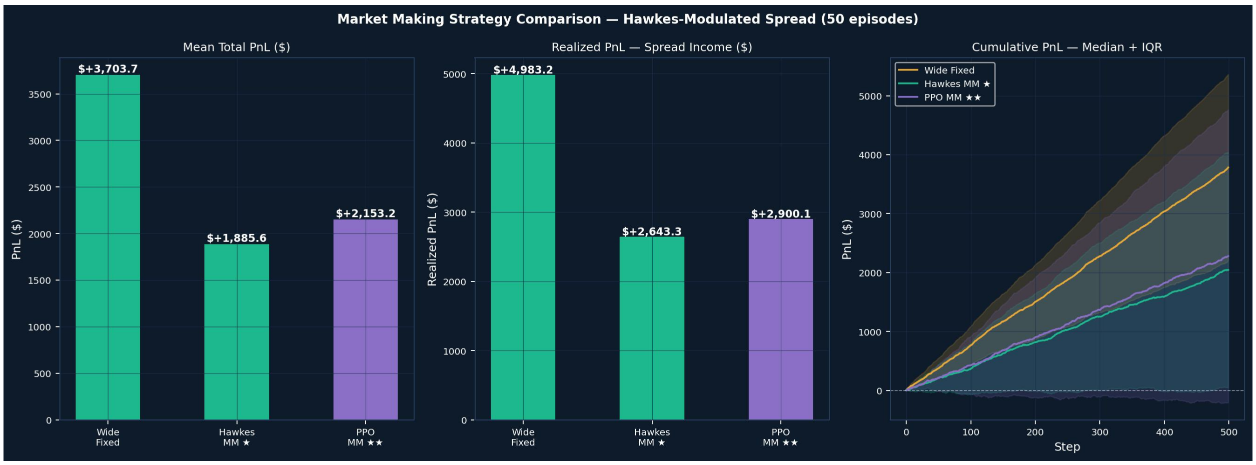


Figure 15: Market making comparison over 50 episodes. Left: cumulative PnL. Centre: realized PnL bar chart. Right: average absolute inventory. PPO MM achieves lowest inventory (2.008, −21% vs. Wide Fixed) with competitive PnL +$2,153.

9. Discussion

9.1 $\lambda(t)$ as a Universal Microstructure Signal

The main finding is that one number—$\lambda(t)$, computed in real time from trade arrival times—helps across four different financial problems. Volatility forecasting, option pricing, delta hedging, and market making all improve when we add $\lambda(t)$. This suggests that Hawkes intensity picks up something real about the market: how fast information is flowing through order flow, which affects volatility, jump risk, hedge ratios, and liquidity at the same time.

9.2 Relation to Classical Stochastic Control

The DRL approaches solve problems that would normally require HJB equations, but those equations have no clean solution under Hawkes dynamics. The Almgren-Chriss (2001) optimal execution problem has a closed-form solution under constant impact, but (55) makes the HJB non-autonomous. The Avellaneda-Stoikov (2008) market-making framework assumes Poisson arrivals; Hawkes order flow introduces state-dependent intensity beyond standard asymptotic methods. DRL gets around this by learning directly from simulated experience.

9.3 Limitations

Data scope. We only use a single 5.38-hour window from one exchange. The Hawkes parameters may change at different times of day or on different exchanges.

Univariate model. A bivariate Hawkes model (**Bacry2015**) would allow buy/sell cross-excitation and permanent vs. transient impact decomposition.

Backtest realism. Simulation environments omit order book depth dynamics, latency, and regulatory constraints.

Stationarity. The branching ratio 0.185 implies stationarity, but BTC flash crashes can transiently push $\alpha/\beta > 1$.

10. Conclusion

This paper built a Hawkes process framework for BTC order flow and demonstrated its applicability across four interconnected domains of quantitative finance. Beginning with the theoretical progression from GBM through Itô's lemma, Merton jump-diffusion, and Lévy processes, we motivated the Hawkes self-exciting process as the natural model for high-frequency cryptocurrency order flow. MLE calibration on 80,000 BTC tick trades yielded a stationary process with branching ratio 0.185 and strong, short-lived self-excitation.

The Hawkes intensity $\lambda(t)$ was applied across four problems:

1. **Hawkes-Vol**: 66.3% directional accuracy in realized volatility forecasting, $R^2 = 0.47$—first model bridging BTC tick microstructure and macro volatility in closed form.

2. **Hawkes+NIG pricing**: ATM IV reduced from 40.25% to 20.40%; left-skewed IV smile reproduced across all tenors.

3. **Hawkes-adjusted delta**: +868% PnL improvement over BS delta (+\$308.5 vs. +\$31.9); PPO agent extends to +\$361.3.

4. **DRL with Hawkes state**: SAC achieves +99.2% vs. TWAP in execution; PPO achieves −21% inventory risk and +\$2,153 PnL in market making.

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